OPPORTUNITY REPORT — FOR TOKIO MARINE

# Logistics Risk Management Worldwide

Value-chain mapping and pain-point analysis across six stages of the freight risk journey — from vendor procurement through cargo insurance and claims

568 evidence-backed claims across 328 unique source domains. Six-stage vertical-slice pipeline with per-stage validation, deep-research remediation, and cross-stage reconciliation. Gate:

SHIP WITH FLAG

EXECUTIVE SUMMARY

# The freight risk journey has three structural gaps that technology has not closed

EVIDENCE CORPUS

**568** claims

| | | |
|---|---|---|
| **13.9%** High confidence (79 claims with primary-source anchor) | **32.9%** Medium confidence (187 claims with corroborated sources) | **53.2%** Low confidence (302 claims — vendor, aggregator, or assumed) |
| **328** Unique source domains across the corpus | **6** Stages independently validated & gated | **10** Deep-research remediations on weak claims |

1 **Subcontracting creates a structural authorization gap.** RTTVPs lose granular signal once a 3PL subcontracts to a non-integrated carrier; fewer than 30% of carrier bases are audited annually; cargo insurers cannot price the subcontracted leg differentially.[1,2,3]

2 **Insurance pricing ignores real-time journey data.** 88% of fleets deploy telematics but only 23% of insurer programs are underwriting-led. Insurance costs rose 12.5% in 2023 and 3.0% in 2024 without differentiation by monitoring investment.[4,5]

3 **The Vesttoo collapse exposed reinsurance verification gaps.** $4B fraudulent LOC exposure, $2.8B confirmed forgeries. Munich Re/Swiss Re cut MGA participation by >50% in 2024, tightening capacity for all cargo InsurTech platforms.[6]

**Tokio Marine relevance**

These three gaps converge at cargo insurance underwriting — the exact point where Tokio Marine's scale, data assets, and distribution network can create differentiated value. The SEA corridor (Indonesia, Thailand, Malaysia) amplifies all three: no FMCSA equivalent, no TAPA-certified parking, and 53–69% of trade-by-value estimated uninsured.

FRAMEWORK

# Six-stage freight risk value chain

WRIGHT PARTNERS

Each stage was researched by a dedicated agent, validated per-stage by a source-validator, and re-anchored by deep-research where claims were weak. A cross-stage validator reconciled inconsistencies.

STAGE 1

### Vendor Procurement & Counterparty Risk

Carrier identity, onboarding, insurance verification, fraud detection. 77 claims, 44 unique domains. The pre-booking gate.

STAGE 2

### TMS, Dispatch & Freight Orchestration

Load assignment, rate confirmation, double-broker prevention. 82 claims, 68 unique domains. Where risk crystallizes into spend.

STAGE 3

### Fleet & Driver Management / Telematics

ELD, dashcams, driver behavior, insurance-grade data. 101 claims, 54 unique domains. The safety-data production layer.

STAGE 4

### In-Transit Visibility & Cargo IoT

RTTVPs, cargo sensors, multimodal tracking. 121 claims, 68 domains. Weakest stage: 69.4% Low, 0 High. Needs field validation.

STAGE 5

### Security, Escort & Physical Protection

Cargo security, secure parking, escort services, theft recovery. 103 claims, 45 unique domains. 21 High-confidence claims.

STAGE 6

### Cargo Insurance, Claims & Incident Response

InsurTech, parametric, claims platforms, incident coordination. 84 claims, 84 unique domains. Highest diversity.

## Per-stage quality metrics

| Stage | Name | Claims | High | Med | Low | Domains | Gate |
|---|---|---|---|---|---|---|---|
| 1 | Vendor Procurement | 77 | 11 | 25 | 41 | 44 | SWF |
| 2 | TMS & Dispatch | 82 | 14 | 24 | 44 | 68 | SWF |
| 3 | Fleet & Telematics | 101 | 14 | 34 | 53 | 54 | SWF |
| 4 | Visibility & Cargo IoT | 121 | 0 | 37 | 84 | 68 | SWF |
| 5 | Security & Escort | 103 | 21 | 39 | 43 | 45 | SWF |
| 6 | Insurance & Claims | 84 | 19 | 28 | 37 | 84 | SWF |

*SWF = Ship With Flag. Low% in 30–70% band across all stages. Five aggregator domains auto-downgraded. Two [VERIFY-N] tags* *require resolution before external use.*

WRIGHT PARTNERS

# Carrier identity fraud is accelerating faster than FMCSA-anchored vetting tools can detect it — and no equivalent vetting infrastructure exists in SEA

| 77 | $725M | 914K | $119B |
|---|---|---|---|
| CLAIMS | US CARGO THEFT 2025 | FRAUD ATTEMPTS BLOCKED (HIGHWAY 2024) | ASEAN ROAD FREIGHT MARKET |

### PP 1.1: Identity fraud at the carrier onboarding layer

In Q1 2026, 50% of freight theft involved carriers with clean records. Highway blocked 527,940 fraudulent emails in Q1 2026 alone (+49.9% YoY). Average loss per incident rose to $273,990 in 2025 (+46% in two years). FMCSA biometric verification halved new-authority approvals — implying half of prior applicants were fraudulent.

FBI IC3 PSA-260430 (Apr 2026); Verisk CargoNet 2025; Highway Q1 2026 Freight Fraud Index; TIA State of Fraud Apr 2025

### PP 1.2: Zero purpose-built counterparty vetting in the $119B ASEAN road freight market

Indonesia Dishub, Thailand DLT, and Malaysia LPKP each maintain carrier records but none exposes a commercial API. All 20 US vetting platforms reviewed have no SEA carrier database. World Bank LPI 2023: Indonesia ranks 61st globally. Manual SEA vetting costs $16–$184 per carrier; a mid-size 3PL incurs $4K–$110K/yr in vetting labor.

Mordor Intelligence ASEAN Road Freight 2025; World Bank LPI 2023; OECD ASEAN Logistics 2021; talent.com & JobStreet SEA salary data 2026

### PP 1.3: Six-signal vetting fragmentation creates compliance fatigue

No single platform consolidates authority status, safety score, insurance, reputation, behavioral fraud, and beneficial ownership. Mid-size brokers pay $340–$500/mo per tool. Insurance data silos are a leading cause of broker-carrier relationship breakdowns in 2025.

RMIS/FreightWaves; Certificial COI benchmarks; Carrier Assure; DAT CarrierWatch

*Sources: Verisk CargoNet (verisk.com); FBI IC3 PSA-260430 (ic3.gov); Highway (highway.com); TIA (tianet.org); Mordor Intelligence; World Bank LPI 2023; OECD 2021*

# Platform automation accelerates dispatch speed but structurally eliminates the identity re-verification step at each tender event

| 82 | 2.15% | $3.8B | $113M |
| --- | --- | --- | --- |
| CLAIMS | FRAUD RATE ON PLATFORMS (2025) | CONVOY PEAK VALUATION BEFORE COLLAPSE | PREVENTED FRAUD (TRIUMPHPAY–HIGHWAY) |

### PP 2.1: Fraud exposure at the booking layer

Identity fraud attempts rose from 0.53% (2023) to 2.15% (2025) — a 306% increase. TMS automation that matches carrier MC numbers to loads does not re-verify identity at dispatch. Average company loss per TIA survey: $402K/yr. No industry-wide dispatch-time re-verification standard exists.

IDScan.net VeriScan (1M+ logistics transactions); TIA Fraud Report 2025; TriumphPay/Highway partnership data

### PP 2.2: Enterprise TMS implementation creates a 12–24 month risk window

66% of large enterprise tech projects end in partial or total failure. Implementation costs $2M–$6M in services on top of $1M–$5M+ annual license. During transition, shippers operate split systems with degraded data quality and no integrated risk enforcement.

Gartner Critical Capabilities for TMS (paywalled); Free Press Journal Dec 2025 ($1M/yr SAP TM)

### PP 2.3: Digital freight platform concentration risk

Convoy's October 2023 collapse stranded 80,000 carriers. $2.6M in unpaid carrier payments; platform sold to DAT for ~$250M (93% discount from $3.8B peak). The IKEA interpleader showed shippers, carriers, and bonding companies with conflicting obligations structured through a broker carrying only a $75K surety bond.

Convoy bankruptcy filings; DAT acquisition; IKEA interpleader ($500K, 58 parties)

### PP 2.4: Freight audit catches billing errors but misses identity fraud

Freight audit catches 3–7% billing errors on $100M spend. But rate-validation logic cannot detect that the MC number presenting an invoice did not physically move the load. TriumphPay–Highway prevented $113.58M in fraud (Jan 2023–2024).

Cass ($38B+ annual audit volume); TriumphPay; National Shippers Strategic Transportation Council

# Telematics generates the behavioral data most predictive of future losses — but the insurance industry does not use it to price risk

| 101 | $4.9B | 88% | 23% |
| --- | --- | --- | --- |
| CLAIMS | COMMERCIAL AUTO UNDERWRITING LOSS (2024) | FLEET TELEMATICS DEPLOYMENT | INSURER PROGRAMS UNDERWRITING-LED |

### PP 3.1: Telematics data is siloed by carrier — shippers and insurers cannot access it

Samsara’s terms prohibit sharing customer data without consent. project44’s integration requires a multi-step manual carrier authorization. For spot-market loads, driver-app authorization completion drops sharply. The result: cargo journeys lose continuous visibility at the exact point where risk is highest.

Samsara ToS; project44 integration guide; Shippeo driver-app documentation

### PP 3.2: 14 consecutive years of commercial auto underwriting losses despite telematics ubiquity

$4.9B underwriting loss in 2024. Insurance premiums hit $0.102/mile (4.5% of operating costs). Nuclear verdicts reached 135 in 2024 (+52%), totaling $31.3B (+116%). 70% of fleet managers with telematics have never been asked by their insurer to share the data.

ATRI 2025 Operational Costs of Trucking; SambaSafety 2025 (n=402); CIAB quarterly P/C Market Surveys

### PP 3.3: No ELD mandate exists anywhere in Southeast Asia

Indonesia’s HoS framework (UU No. 22/2009) has no digital audit mechanism. Thailand reversed its mandatory GPS requirement. Vietnam has no digital tachograph standard. SEA carrier safety compliance rests on unenforced paper records. Indonesia: ~27,000 road fatalities/yr.

Indonesia UU No. 22/2009; WHO Global Status Report on Road Safety; Mordor Intelligence SEA telematics

### PP 3.4: Alert fatigue undermines video telematics safety ROI

Uncalibrated systems require 30–60 min/day sorting false triggers. At one safety manager per 150–300 vehicles, the alert volume defeats the coaching objective. Aggregate cost: $100K–$280K/yr for 20-fleet enterprises.

SureCam alert fatigue documentation; Lytx DriveCam; Motive AI Safety

WRIGHT PARTNERS

# RTTVP carrier networks market breadth but fewer than 2% of connected carriers meet preferred data-quality standards

| | | | |
|---|---|---|---|
| **121** | **0** | **$6.3B** | **13.8M** |
| CLAIMS | HIGH-CONFIDENCE CLAIMS (WEAKEST STAGE) | VISIBILITY PLATFORM MARKET (2024) | CARGO TRACKING DEVICES INSTALLED |

### PP 4.1: Visibility quality is structurally tiered — spot loads fall into dark zones

project44's Preferred Carrier program covers 3,580 carriers (~2% of its 245K+ network). Class 3 phone-app tracking depends on driver action; for spot-market loads, completion drops sharply. Shippeo confirms: if the subcontractor isn't already connected, "there can be no real-time tracking."

project44 Preferred Carrier Program 2024; Shippeo blog; Gartner RTTVP MQ 2024

### PP 4.2: Cargo IoT sensors are point-deployed by a single party — data does not cross the chain

Tive's "Shared Shipments" requires manual link generation per third party. Parsyl's Data Partner Program covers only its own policyholders. A pharma pallet moving through 4 entities has native access for only the account holder. 13.8M devices installed globally but only Tive has deployed 2M total.

Berg Insight Cargo Tracking 2024; Tive & Parsyl product documentation; Controlant cold chain

### PP 4.3: Multimodal visibility degrades at every modal interface

Average mid-to-large shipper runs 5–10 separate visibility tools. At the port gate, rail yard, and air cargo terminal, there is no continuous signal — dwell periods are exactly where cargo is most physically vulnerable. Flexport's system was documented generating erroneous data at port handoffs.

Independent freight-tech analysis (FreightWaves); project44/FourKites/Wakeo multimodal claims

**Caution: Stage 4 is the weakest in the corpus**

69.4% Low confidence, zero High-confidence claims. The in-transit visibility category most needs field validation before commercial use. project44 revenue figures carry a [VERIFY-1] tag.

# Post-theft response coordination is manual, sequential, and consumes the 24–48 hour recovery window — the cargo categories most needing physical protection are those for which escort is least economically viable

| | | | |
|---|---|---|---|
| **103**<br>CLAIMS | **21**<br>HIGH-CONFIDENCE CLAIMS | **~10%**<br>ARREST RATE FOR CARGO THEFT (US) | **78K**<br>SECURE TRUCK PARKING SHORTAGE (US) |

**PP 5.1: No common incident-response workflow across shipper, 3PL, carrier, insurer, and law enforcement**

$725M in US/Canada cargo theft in 2025 (+60% YoY). Average: $273,990/incident. Arrest rate ~1 in 10. CargoNet captured 787 Q1 2025 incidents vs. Overhaul's 505 — demonstrating fragmented, non-overlapping intelligence. FBI PSA260430 documents cyber-enabled tactics with extended undetected access.

Verisk CargoNet 2025; FBI IC3 PSA-260430; AAR/CORCA; Overhaul Security Index

**PP 5.2: Secure truck parking deficit — 40% of thefts at unsecured roadside locations**

FHWA documents a 78,000-space shortage; two-thirds of rest events at unauthorized locations. TAPA data: 137 thefts at unclassified parking vs. 3 at TAPA-certified — a 46:1 ratio. In SEA, no TAPA PSR-equivalent exists on major freight corridors.

Munich Re 2025 Cargo Theft Report; FHWA Truck Parking; TAPA EMEA; ATRI parking surveys

**PP 5.3: Physical escort pricing creates a cost-value mismatch for high-theft commodity categories**

Brazil's road cargo companies spend 14% of revenue on security (NTC&Logistíca Feb 2025). Food/beverage is the #1 stolen commodity (22% of incidents) but the category where per-unit escort cost is least justified. In SEA, no regional equivalent to Brink's/Prosegur operates on inland corridors.

NTC&Logistíca (9th CNT Forum Feb 2025); BSI/TT Club 2024; ReCAAP ISC 2025

**PP 5.4: Cargo theft intelligence is fragmented across competing private platforms**

CargoNet, SensiGuard, and Overhaul report non-overlapping incident sets. WTW 2026 Guide identifies eight emerging risk categories. No single platform maps all eight. Small carriers learn about new tactics months late — typically through a loss, not a warning.

WTW 2026 Strategic Cargo Theft Guide; Overhaul Intelligence; CargoNet; NICB

# Cargo insurance pricing remains anchored to declared values and pooled actuarial tables — the Vesttoo collapse exposed structural collateral-verification risks across the MGA ecosystem

| 84 | 84 | 19 | $22.6B |
|---|---|---|---|
| CLAIMS | UNIQUE DOMAINS (HIGHEST DIVERSITY) | HIGH-CONFIDENCE CLAIMS | GLOBAL CARGO PREMIUMS (IUMI 2024) |

**PP 6.1: Pricing ignores per-shipment journey data despite telematics ubiquity**

88% telematics deployment but only 23% of insurer programs underwriting-led (SambaSafety n=402). 79% of non-sharing fleets say no insurer asked for data. ATRI: insurance hit $0.102/mile (+12.5% in 2023). Red Sea crisis: war-risk premiums jumped 2,700% while bordereaux reporting lagged by weeks.

SambaSafety 2025 Telematics Report (primary, n=402); ATRI 2025; CIAB quarterly surveys; Carrier Mgmt/Hercules Systems Apr 2026

**PP 6.2: Claims resolution takes 30–90 days because loss documentation does not exist in machine-readable form**

Proving loss requires BOL, POD, packing list, inspection report, photos from multiple parties in paper/PDF form. Clean delivery receipt is the #1 denial trigger. Only 30% of carrier bases audited annually. Loadsure/Breeze achieve same-day payout for clean claims but 30–40% of claims involve disputed documentation.

Insurance Edge Apr 2026; Loadsure/Google Cloud Document AI; Carmack Amendment timeline; University of Tennessee supply chain research

**PP 6.3: Parametric cargo insurance has not scaled beyond cold-chain specialty niches**

Parsyl ColdCover: 97% claims reduction for vaccine cold chains. But for general cargo, no parametric trigger has sufficiently low basis risk. IAIS/FSI Dec 2024 identifies basis risk as primary adoption barrier. SEA regulators (OJK, OIC) lack specific parametric product frameworks.

Parsyl ColdCover; IAIS/FSI Dec 2024 joint note; Singapore MAS InsurTech guidance

**PP 6.4: Vesttoo collapse — $4B in fraudulent reinsurance capacity**

$2.8B confirmed LOC forgeries. Markel $127.75M exposure. Aon $197M provision. Munich Re/Swiss Re cut MGA participation >50% in 2024. All specialty-line cargo MGAs face higher reinsurance costs and stricter fronting-carrier oversight.

Artemis ILS; AM Best; Browne Jacobson; Pinnacle Actuarial; ALIRT Insurance Research

PRIORITIZATION

# Three waves of opportunity for Tokio Marine

Sequenced by evidence strength, Tokio Marine's existing capabilities, and SEA market timing.

WAVE 1 — 0–6 MONTHS

### Telematics-linked underwriting

**Opportunity:** Close the gap between 88% fleet telematics deployment and 23% underwriting-led insurer programs. Tokio Marine can differentiate cargo and commercial auto pricing by ingesting telematics data that 79% of fleets say no insurer has asked for.

**Evidence:** High confidence. SambaSafety primary survey (n=402). ATRI primary data. 14 consecutive years of commercial auto losses ($4.9B in 2024).

**Action:** Pilot with 3–5 large fleets in Japan/SEA. Integrate Samsara/Motive API feeds into underwriting workflow.

WAVE 2 — 6–18 MONTHS

### SEA carrier-risk data layer

**Opportunity:** Build or acquire the first machine-readable carrier-risk database for ASEAN road freight — the "FMCSA SAFER for SEA." $119B market with zero purpose-built counterparty tooling. Tokio Marine's SEA presence (TM Indonesia, MSIG) provides distribution.

**Evidence:** Medium confidence. Mordor Intelligence market sizing. World Bank LPI. No Tier-1 primary source for ASEAN road freight market size.

**Action:** Partner with Dishub/DLT for data feeds. Pilot in Indonesia domestic road freight.

WAVE 3 — 12–24 MONTHS

### Per-shipment parametric cargo for SEA corridors

**Opportunity:** Deploy parametric cargo insurance on SEA cold-chain corridors where IoT data availability and high spoilage rates create low basis risk. Expand Parsyl-style ColdCover model to Indonesia–Singapore pharma and food corridors.

**Evidence:** Medium confidence for cold chain. Low for general cargo (basis risk). IAIS/FSI regulatory guidance exists. OJK/OIC need specific parametric frameworks.

**Action:** Regulatory engagement with OJK (Indonesia) and MAS (Singapore). Pilot on pharma cold chain with Tive/Sensitech sensor partners.

## Top 3 questions for field validation

1. **SEA sub-carrier cargo coverage:** Interview cargo insurance brokers in Jakarta and Bangkok on whether sub-contracted road legs are actually covered under shipper or 3PL open cargo policies. Resolves ASSUMED-3, ASSUMED-4, ASSUMED-5.

2. **Insurer–telematics integration in SEA:** Interview a marine cargo underwriter at Tokio Marine Indonesia or MSIG on whether they verify monitoring levels when setting rates on road freight legs. Resolves ASSUMED-2.

3. **IoT cold-chain deployment on Indonesia domestic road:** Interview DHL Indonesia or Kerry Logistics on what fraction of domestic temperature-controlled legs have IoT sensor coverage vs. paper-log methods. Resolves ASSUMED-8.

# Sources 1–12

| # | Publisher / Title | Date | URL | Supports |
|---|---|---|---|---|
| 1 | Verisk CargoNet — Annual Cargo Theft Analysis | 2025 | cargonet.com | $725M theft losses; $273,990/incident; 60% YoY increase |
| 2 | FBI IC3 PSA-260430 | Apr 2026 | ic3.gov/PSA/2026 | Confirms CargoNet 2025 figures; fictitious pickups as dominant driver |
| 3 | TIA — State of Fraud in the Industry | Apr 2025 | tianet.org | $402K avg. company loss; 83% experienced 3+ fraud types |
| 4 | Highway — Q1 2026 Freight Fraud Index | May 2026 | globenewswire.com | 50% of Q1 2026 theft involved clean-record carriers |
| 5 | SambaSafety — 2025 Telematics Report | 2025 | sambasafety.com | 88% telematics deployment; 23% underwriting-led; n=402 primary survey |
| 6 | ATRI — Operational Costs of Trucking | 2025 | truckingresearch.org | $0.102/mile insurance; +12.5% in 2023; +3.0% in 2024 |
| 7 | CIAB — Quarterly P/C Market Surveys | 2022–2024 | ciab.com | 8–12% commercial auto premium increases |
| 8 | BSI / TT Club — 2024 Cargo Theft Report | Apr 2025 | ttclub.com | 41% theft during transit; 22% food/beverage; SEA risk profile |
| 9 | World Bank — Logistics Performance Index | 2023 | lpi.worldbank.org | Indonesia 61st; Singapore 1st; ASEAN LPI spread |
| 10 | Mordor Intelligence — ASEAN Road Freight Transport Market | 2025 | mordorintelligence.com | $119.43B market size; 6.04% CAGR |
| 11 | IUMI — Global Marine Insurance Report | 2024 | iumi.com | $22.64B global cargo premiums; triangulation for uninsured share |
| 12 | ADB — Trade Finance Gaps, Growth, and Jobs Survey | 2023 | adb.org | 45% SME application rejection; $2.5T unmet demand; proxy for uninsured SME |

# Sources 13–24

| # | Publisher / Title | Date | URL | Supports |
|---|---|---|---|---|
| 13 | Munich Re — Cargo Theft Report | 2025 | munichre.com | 40% of theft at unsecured roadside parking |
| 14 | TAPA EMEA — Cargo Crime Intelligence | 2024 | tapaemea.org | 137 thefts at unclassified vs. 3 at certified parking (46:1 ratio) |
| 15 | ReCAAP ISC — Annual Report | 2025 | recaap.org | SOMS piracy data; SEA maritime security baseline |
| 16 | NTC&Logistíca — 9th CNT Public Security Forum | Feb 2025 | ntcelogistica.org.br | 14% of revenue on cargo security in Brazil |
| 17 | Insurance Edge — Cargo Subcontracting Analysis | Apr 2026 | insurance-edge.net | 30% carrier bases audited annually; subcontracted-leg liability gap |
| 18 | Artemis — Vesttoo ILS Coverage | 2023–2024 | artemis.bm | $4B ILS capacity; $2.8B confirmed forgeries |
| 19 | IAIS / FSI — Parametric Insurance Joint Note | Dec 2024 | iaisweb.org | Basis risk as primary adoption barrier; regulatory frameworks |
| 20 | Berg Insight — Cargo & Container Tracking | 2024 | berginsight.com | 13.8M installed tracking devices globally |
| 21 | WTW — Guide to Strategic Cargo Theft | 2026 | wtwco.com | 8 emerging risk categories; cyber-enabled theft tactics |
| 22 | FHWA — Truck Parking Survey & Data | 2023–2024 | fhwa.dot.gov | 78,000-space shortage; 2/3 of rest events at unauthorized locations |
| 23 | OECD — Competition Assessment: Logistics in ASEAN | 2021 | oecd.org | Regulatory fragmentation as logistics barrier; cross-border carrier permit gaps |
| 24 | Gartner — Magic Quadrant for RTTVPs | 2024 | gartner.com | 7 evaluated vendors; RTTVP market definition and competitive landscape |

**Full corpus provenance**

568 numbered claims drawn from 328 unique domains across 6 stages. Five aggregator domains (Mordor Intelligence, MarketsAndMarkets, Grand View Research, GMInsights, Precedence Research) were auto-downgraded to Low confidence. Two [VERIFY-N] tags and 10 [ASSUMED-N] tags remain for resolution. Complete per-claim source mapping available in the raw research artifacts.

NEXT STEPS

## Contact Information

We deliver ventures, not slides. This research identifies three actionable waves for Tokio Marine to deploy capital and capability against verified gaps in the global logistics risk value chain. Let's discuss next steps.


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